

Literature Review

Due Date: **Week 5 (2/13)** on Canvas.

Points: 20 points

Description: This worksheet will aid you in writing your paper by assisting you in your search of the literature to

1. find and summarize relevant previous studies
2. evaluate data results to describe patterns in the research as they relate to your research questions

Conducting a literature review (aka reading papers!) is time-consuming and not possible to complete in its entirety in class. Researchers spend a lot of time reading and taking careful notes on the studies they find as they gather information to write their own research papers. You are asked to do the same in this homework assignment; please take time outside of class to search for articles and to evaluate previous findings, with the goal of connecting your thoughts to our experiment.

A portion of this literature review worksheet is structured as a tutorial with our Library Research Guide (on Canvas).

Instructions: **Conduct** a collaborative literature review on intellectual humility, victims' post-transgression responses, and apology.

Section 1 of this literature review worksheet is structured as an **in-class tutorial** with our Library Research Guide (on Canvas). This tutorial helps you begin your search by identifying the concepts and keywords of interest and by finding your first five potential journal article citations.

Section 2 requires additional time **outside of class** and will help you evaluate your chosen articles for their "fit" with our class experiment (i.e., potential to reference in your introduction of your paper to make a given argument about your experiment). This will be *extremely* helpful to you as you prepare to write your research report.

- - - - -
Section 1: Questions to answer while exploring our RM Lab Library Research Guide

A) **Draft initial hypothesis:** Use your own words to draft a hypothesis for our class experiment that will investigate the impact of intellectual humility (vs. control) and victims' post-transgression response (forgiving vs. unforgiving) on apology behavior. *Tip: remember the 3 key components from lecture.*

Do not worry about writing the "perfect" answer now.

Hypothesis 1: How will apologies differ between the IH and control conditions?

If the participant is assigned to the IH condition, then their apologies will be more effective than the control group because they are more likely to admit wrong-doing.]

Hypothesis 2: How will apologies differ between the forgiving and unforgiving conditions?

If the participant is assigned to the high forgives condition, then their apologies will be more effective than the unforgiving condition because they will feel less guilt about their mistake.

B) **Identify 5 search terms/concepts:** Please identify 5 search terms that describe the research area and topics for our class experiment. Identify keywords and/or discipline-specific terminology to use to locate research for your paper. *Tip: you can use quotation marks, operators such as AND OR, and wild cards such as * to help narrow your search.*

1. "Intellectual humility" and "conflict"
2. "Intellectual humility" and "forgiv*"
3. "Intellectual humility" and "reconcile"
4. "Intellectual humility" and "empath*"
5. "Intellectual humility" and "receptive*"

C) **Find two (2) research articles:** Using the search terms you have identified in B, find two (2) research papers to use as background reading for your paper. Read the abstracts of these papers and think about how

these studies compare to our class experiment. Copy the APA-formatted reference and the abstract for your reference.

Use the table on the page 4 to copy and paste the reference and abstract you want to keep.

D) Can you think of a hypothesis for the interaction? How would both IV #1 and IV #2 combined impact apology behavior?

Participants in the Intellectual humility and forgiving conditions will have the highest quality apologies across all conditions.

E) **Find two additional research articles (to make at least 6 total, including the studies already included in the table):** Using some of the first articles you've found, branch out your literature review to find three (2) additional articles you can add to the next page.

Copy and paste the references and abstracts to the table on page 4.

List of the articles you have identified

#	Reference (APA style)	Abstract
1: Defining IH	Porter, T., Elnakouri, A., Meyers, E. A., Shibayama, T., Jayawickreme, E., & Grossmann, I. (2022). Predictors and consequences of intellectual humility. <i>Nature Reviews Psychology</i> , 1(9), 524-536. https://doi.org/10.1038/s44159-022-00081-9	In a time of societal acrimony, psychological scientists have turned to a possible antidote — intellectual humility. Interest in intellectual humility comes from diverse research areas, including researchers studying leadership and organizational behaviour, personality science, positive psychology, judgement and decision-making, education, culture, and intergroup and interpersonal relationships. In this Review, we synthesize empirical approaches to the study of intellectual humility. We critically examine diverse approaches to defining and measuring intellectual humility and identify the common element: a meta-cognitive ability to recognize the limitations of one's beliefs and knowledge. After reviewing the validity of different measurement approaches, we highlight factors that influence intellectual humility, from relationship security to social coordination. Furthermore, we review empirical evidence concerning the benefits and drawbacks of intellectual humility for personal decision-making, interpersonal relationships, scientific enterprise and society writ large. We conclude by outlining initial attempts to boost intellectual humility, foreshadowing possible scalable interventions that can turn intellectual humility into a core interpersonal, institutional and cultural value.
2: IH and Apologies	Ludwig, J. M., Schumann, K., & Porter, T. (2022). Humble and apologetic? Predicting apology quality with intellectual and general humility. <i>Personality and Individual Differences</i> , 188, 111477. https://doi.org/10.1016/j.paid.2021.111477	Apologies are powerful predictors of reconciliation, but transgressors often fail to offer optimal, high-quality apologies that are comprehensive and non-defensive. We tested whether intellectual humility and general humility predict the use of high-quality apologies versus taking no action to resolve a conflict, and the processes that mediate these associations using online vignette experiments. In Study 1 (N = 397), transgressors with greater intellectual humility offered higher-quality apologies and were less likely to take no action following a relational offense. However, these associations did not remain significant when controlling for general humility. In Study 2 (N = 394), intellectual humility uniquely predicted greater apology comprehensiveness and less inaction following an intellect-based offense, demonstrating its context-specific associations with apology behavior. By contrast, general humility was a robust predictor of higher-quality apologies

		and less inaction across offense contexts. Consistent with recent theorizing on psychological barriers to apologizing, both studies also found support for the mediating roles of empathic effort and self-protection.
3: at least another study related to IH and conflict behaviors	Koetke, J., Schumann, K., Welker, K., & Coleman, P. T. (2024). Intellectual humility is reliably associated with constructive responses to conflict. <i>PLoS ONE</i> , 19(9), 12. https://doi.org/10.1371/journal.pone.0309848	Conflict is a ubiquitous, but potentially destructive, feature of social life. In the current research, we argue that intellectual humility—the awareness of one’s intellectual fallibility—plays an important role in promoting constructive responses and decreasing destructive responses to conflict in different contexts. In Study 1, we examine the role of intellectual humility in interpersonal conflicts with friends and family members. In Study 2, we extend this finding to workplace conflicts. In both studies we find that intellectual humility predicts more constructive and less destructive responses to conflict. This work extends the burgeoning literature on the benefits of intellectual humility by demonstrating its association with responses that help defuse conflictual encounters.
4: at least another study related to victims’ post-transgression responses and apology behavior	Struthers, C. W., Eaton, J., Shirvani, N., Georghiou, M., & Edell, E. (2008). The Effect of Preemptive Forgiveness and a Transgressor’s Responsibility on Shame, Motivation to Reconcile, and Repentance. <i>Basic and Applied Social Psychology</i> , 30(2), 130–141. https://doi.org/10.1080/01973530802209178	The primary aim of this research was to examine the effects of an injured party's preemptive forgiveness (grudge, none, explicit, implied) on a transgressor's repentance. We also explored the moderating role of a transgressor's judgment of responsibility (low, high) and the mediating role that feelings of shame and motivation to reconcile play in the preemptive forgiveness → repentance process. The results of two experiments showed that different types of forgiveness had different effects on repentance and prosocial conduct. In Study 2, we extended these findings by examining the moderating role that judgment of responsibility plays and the mediating role that shame and motivation to reconcile plays in the preemptive forgiveness → repentance process. These experiments show that, relative to holding a grudge, no forgiveness, and explicit forgiveness, the most effective preemptive prosocial strategy was implied forgiveness, particularly when judgments of responsibility were low. In addition, a stepdown analysis suggests that the effects of forgiveness and responsibility on repentance are mediated by a transgressor's shame and motivation to reconcile.

5: Your choice	Dmytro O. Rebrov, C. Ward Struthers, Jingyuan Li, Ariel Shoikhedbrod, Joshua R. Guilfoyle, (2022). The moderating role of transgressors' apology on the association between the Dark Triad and victims' post-transgression responses, <i>Personality and Individual Differences</i>, Volume 194, https://doi.org/10.1016/j.paid.2022.111639.	The goal of this research was to examine the association between victims' Dark Triad personality and their social motivation to not forgive (unforgiveness) or forgive transgressors after an interpersonal conflict. The moderating role of transgressors' apology as well as the mediational role of victims' indignation to explain the moderated relation was also tested. Unforgiving responses (operationalized as revenge and avoidance) were predicted to be stronger, and forgiving responses, weaker, among those exhibiting characteristics consistent with the Dark Triad. We also predicted a counterintuitive, but theoretically consistent, stronger positive relation between the Dark Triad and unforgiving responses when victims received an apology. In addition, we predicted that indignation would explain this moderated relation. Two studies using an adult community and an undergraduate sample, respectively, confirmed the moderation hypothesis (Studies 1 and 2) but not the mediated moderation hypothesis (Study 2).

SECTION 2: In-depth evaluation

Now you have identified at least 5 papers for potential use in your Introduction and/or Discussion of your paper, you will need to evaluate these references carefully.

Are they really related to our research?

Are they a good “fit”?

Can you make a strong, convincing argument with them?

Carefully **read each entire paper** and **complete the table** below. This will help you develop an argument to motivate our study. Taking careful notes in the table below will also save you time re-reading the papers when you write your research report.

Complete this activity at home on your own time with your research partner. There may be times in class when your instructor provides time to work on ideas pulled from this worksheet; however, most reading and taking notes will occur outside of the class period. Please describe the studies in your own words.

Note: Use the same numbers for each study as above (to match the references later)

#	What did they do and how?	What was the main result?	How does this relate to our study? (Confirm? Disagree? Expand?)
---	---------------------------	---------------------------	---

1	This paper was an examination of intellectual humility with the goal of defining the construct. This was done by looking at a variety of different definitions across several measures to synthesize a general definition. The methods that were used to produce definitions ranged from self-reported questionnaires to behavioral tasks. The authors also identified threats to intellectual humility including cultural, interpersonal, personal and overlapping variables. Lastly they outlined the importance of intellectual humility by investigating the correlation between different variables such as social, personality and cognitive traits. These correlations were pulled from previous literature on the respective topics.	The future directions in this study found were to further explore intellectual humility's position as "too much of a good thing". Additionally, they ask about more research on how intellectual humility varies from individual to individual, and from culture to culture. Lastly, the study recognizes that intellectual humility, although seen as a positive trait, is not a remedy for all of societal challenges, and remains essentially a tool in the emotional toolbox .	This paper primarily helps us operationalize intellectual humility and related terms. This study looks at a variety of other studies to identify the core traits of intellectual humility which is vital in the process of creating a hypothesis and conducting a effective study. Additionally, this paper looks at the related variables, notably empathy and forgiveness being two of them, something that we are also identifying in our study as a variables of interest. Lastly, this study allows us to build onto a body of research that is well organized and has strong indications for being beneficial to the greater societal context. Using this study, especially in our introduction, would be highly useful in not only defining our terms but also giving reasoning for why this work is so I important and referencing the further directions section as we are building off of their requests.
---	---	---	--

2	The authors looked at apology quality and how this can be predicted by a transgressor's humility (either general humility or intellectual humility). There were two studies conducted. In the first study, participants took several different self-reported questionnaires. This included: an intellectual humility scale, a general humility scale, five personality scales, the Brief Big 5 Inventory, the Rosenberg Self-Esteem Scale, the Narcissism Personality Inventory-13, the Need for Cognition scale, the Marlowe-Crowne Social Desirability Scale, an offense vignette, a pseudo self-forgiveness scale, an empathy scale, and an apology scale. In study two, a new batch of participants was sampled. These participants took several self-reported scales, including: the same intellectual humility and general humility scales in study one, an offense severity scale, the same self-protection scale and empathy scale in study one, and a post-offense response scale.	Study 1: The researchers found that apology quality can be strongly predicted by a person's general humility. Study 2: In the context of an intellectual offense, the researchers found that intellectual humility can strongly predict apology behavior. Overall: The researchers concluded that general humility was a better predictor of apology behavior across all offense contexts, not just intellectual offenses.	This paper directly examines the relationship between intellectual humility and apology quality. This is exactly what we are researching in our paper. In addition, this paper looks at general humility. While general humility is not a variable in our study, the results found in this paper that link general humility to higher quality apologies across ALL offense conditions can help explain the findings in our study.
---	--	--	--

3	The researchers conducted two studies to measure how responding constructively in an interpersonal conflict can be predicted by a person's intellectual humility. Study one collected participants from ten different courses across three universities (University of North Texas, Crafton Hills College, and the University of Maryland). These participants were then tasked with completing the following: the GIHS, the CIHS, a scale that measured positive informing, positive evading, positive opening, positive uniting, negative attacking, and negative evading, a social desirability scale. Study two expanded upon study one by looking into workplace conflict. The participants used in this study were all employed by the Government Finance Officers Organization. These participants completed every scale given in study one, as well as a scale measuring conflicts responses with a supervisor.	Study 1: More positive behaviors and less negative behaviors were both predicted by the GIHS and CIHS. Study 2: More positive behaviors in the workplace were predicted by the GIHS and the CIHS. Less negative behaviors were only significantly predicted by the CIHS. Overall: The GIHS was a great predictor of positive behaviors across both the workplace and family/friends. The researchers also concluded that the CIHS was better at predicting negative behaviors in	Our study specifically looks at a situation with a classmate. The paper here seeks to determine conflict responses in situations that involve coworkers, supervisors, family members, and friends. The results found in this study will help us build upon the general idea that intellectual humility promotes higher quality apologies across many different situations. This is especially useful considering they found a correlation in four different interpersonal areas, while our study focuses on just one.
---	--	--	--

4	They conducted two studies. The first was an experimental design conducted in a lab to measure how different types of pre-emptive forgiveness impacted the offenders repentance. They did this by creating a fake tutor scenario, where the participants gave the “student” an incorrect answer which prevented the student from being able to complete their assignment. The “student” then gave one of four different pre-emptive forgiveness responses and the participant’s reactions were recorded. The second study was correlational, and used vignettes to gauge the relationship between preemptive forgiveness and transgressors level of responsibility. This was achieved by having participants imagine they were in a scenario where a coworker was harmed by their behavior, either in a low responsibility condition or high responsibility condition.	The results of study one found that when the victim held a grudge (no pre-emptive forgiveness) they were the least likely to receive an apology. When pre-emptive forgiveness was implied it resulted in the most repentance, even more so than the explicit forgiveness condition. In the second study, the results found that when individuals had low responsibility, and high pre-emptive forgiveness there was the highest likelihood of apology. The results indicated the strongest relationship when both conditions were met, however there was also some relationship with each variable individually.	This relates to our study in terms of looking at post-transgression behavior. This study gives us more information on what makes individuals more or less likely to apologize. Additionally, these studies both focused on the victim’s response which is a significant part of our study. Our hypothesis is that the more forgiving the victim is the more likely they are to receive a high quality apology, and this study reenforces that prediction because the participants in the preemptive forgiveness (study two) and the implicit forgiveness (study one) groups were more likely to apologize. Using the information these studies concluded, we can build on not only what makes someone more likely to apologize, but also what makes someone more likely to deliver a high quality apology by introducing the intellectual humility variable in our procedure.
---	---	---	--

5	This paper also looked at two studies. The first study conducted was a questionnaire delivered to individuals which asked them about their personality traits in relation to the Dark Triad, as well as a conflict that was left unresolved. The participants then reported their post-transgressional response to gauge how they reacted to the conflict they experienced. In the second study had a similar procedure where participants were asked questions about their personality and to describe an unresolved conflict where they were the victim. They were also asked specifically about their level of indignation towards the aggressor. Lastly they were also measured on the same post-transgression scale as the previous paper.	For the first study, they found a significant difference between the level of unforgiveness (TRIM-12) in the group that was rated low on Dark Triad traits. Those who received an apology ranked lower on the TRIM-12 (meaning more forgiving) than those who did not receive an apology. In contrast, there was almost no difference in the group that ranked high in the DT characteristics, with the level of unforgiveness being almost the same, and higher in both cases than the other group. In the second study the results were similar, the only difference being the level of unforgiveness was nearly the same in	Although our research is not explicitly looking at the Dark Triad and it's traits, this study does give us some valuable information about post-transgressional response in relation to apologies. In addition, some of the variables looked at in these studies such as narcissism and psychopathy can be applied to our research in terms of empathy and general humility.
---	---	--	---

You now have everything you need to start working on brainstorming the organization of your paper writing. To get you started on drafting your introduction outline, answer the following two questions for yourself.

Note: You can use the answers to these questions directly in your outline and introduction draft.

1. Looking at the table you created, describe the patterns that you found. Consider the following questions: What outcome occurred the most often? How should we form our hypotheses based on prior research? Reformulate your hypothesis again (see your own drafts in questions A) and D) above.

Throughout all the above studies, there is a consistent trend that forgiveness improves the quality of an apology and reduces the tension of a post-transgression response. As a result of this work, we will retain our hypothesis that the forgiving condition group will be able to deliver a higher quality apology than the nonforgiving condition.

2. What is the gap in the literature that our class's experiment might address? Why is our experiment a necessary next step (e.g., why is it important to look at framing with our second independent variable)? Look for disagreements in the literature, or for areas that have not been explored yet (e.g., specific population, ages, circumstances, cultures, ...).

The predominant issue seems to be the application of post-transgression response mediation. In other words, there needs to be a way to successfully improve the levels of intellectual humility and high quality apologies on a general scale. If we are able to successfully increase intellectual humility levels within our study, it may have significant applications. Additionally, if we find strong correlations between increased intellectual humility and other manipulatable variables, we may also be able to improve in this domain.

